

Factors Influencing Hospital Readmissions: A Health Data Analysis Project

Completed in
August 2025,
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Executive summary

This report analyzes hospital readmissions and identifies the key factors that influence whether patients are readmitted after discharge. Using a dataset of 10,000 patient records, the analysis examines demographic, clinical, and hospital-related variables to uncover patterns associated with readmission.

Statistical analyses were conducted in Python using descriptive statistics, t-tests, and logistic regression modeling to determine which factors most strongly predict readmission.

Key findings show that length of stay (LOS) alone does not significantly predict readmission, while patient-specific factors—such as age and underlying disease—play a much stronger role. Logistic regression analysis revealed that conditions like asthma and influenza, as well as older age, were significantly associated with a higher likelihood of readmission. These insights highlight opportunities for targeted interventions and data-informed discharge planning to reduce preventable readmissions.

Conclusions

The analysis demonstrates that relying solely on hospital metrics such as length of stay is insufficient to predict patient readmission risk. Instead, patient-specific characteristics—especially age and chronic diseases—are stronger indicators of readmission likelihood. These findings suggest that more individualized discharge planning and follow-up care may be required to address the needs of high-risk patients.

Additionally, integrating predictive analytics into hospital workflows could help identify vulnerable patients earlier, allowing for targeted interventions that improve outcomes and reduce healthcare costs.

Recommendations

- **Implement predictive tools** that combine patient demographic and clinical data to identify high-risk individuals prior to discharge.
- **Enhance transitional care programs**, especially for older adults and patients with chronic respiratory illnesses.
- **Educate patients and caregivers** on post-discharge symptom monitoring and when to seek follow-up care.

Introduction

As a nurse, I witness firsthand how hospital readmissions represent a critical challenge in healthcare systems, leading to increased costs, resource strain, and a significant burden for patients and families. Historically, hospitals have relied on measures such as length of stay and discharge status as proxies to predict the likelihood of readmission. However, with the growth of electronic health records and advancements in healthcare data analytics, there is now an opportunity to explore more granular predictors, including patient demographics and clinical characteristics.

The purpose of this report is to analyze patient and hospital factors that influence readmission rates, using a dataset of 10,000 patient encounters. By identifying key predictors of readmissions, this analysis aims to provide actionable insights for hospital administrators and clinicians seeking to improve care transitions, optimize resource use, and ultimately reduce preventable readmissions.

Significance

Understanding the factors that contribute to hospital readmissions enables healthcare providers to identify high-risk patients, tailor discharge planning, and allocate resources more effectively. Focusing on patient-specific characteristics and disease profiles can lead to more equitable and efficient care delivery, ultimately improving patient outcomes while reducing costs associated with preventable readmissions.

Methods and Data

Data Set Overview

The dataset used for this analysis was obtained from [Kaggle's Hospital Data for Patient Readmission Prediction](#). It contains 10,000 patient records with demographic, clinical, and hospital-related variables.

Sample characteristics:

- Age range: 0–95 years (mean $\approx$ 47 years)
- Gender: Male and Female
- Diseases: 30+ conditions, including chronic and acute illnesses
- Setting: Single hospital system (John Hopkins Hospital, multiple units)

Techniques:

Data preprocessing and analysis were conducted in **Python** using *Pandas*, *Seaborn*, and *Matplotlib*.

- Exploratory Data Analysis (EDA): Descriptive statistics, histograms, and bar charts to visualize readmission patterns.

Statistical tests:

- T-tests comparing average length of stay between readmitted and non-readmitted patients.
- Logistic regression modeling to quantify the effect of variables such as age, disease type, length of stay, and discharge status on readmission likelihood.

Data Visualizations:

- Data visualizations supported interpretation of trends and patterns throughout the analysis.

Data workflow:

The analysis began with data cleaning and selection of relevant columns to handle missing values. Descriptive statistics and visualizations explored patterns in readmissions across different patient groups. Key variables—length of stay, age, disease type, and discharge status—were examined using T-tests and logistic regression to identify the strongest predictors of readmission.

Limitations and constraints

This analysis was limited by the dataset's single-hospital scope, which may not represent broader populations or different healthcare systems. Additionally, the dataset lacked socioeconomic and behavioral factors, which could further influence readmission risk. Future analyses incorporating multi-center data and additional variables would enhance model generalizability.

Findings and Discussion

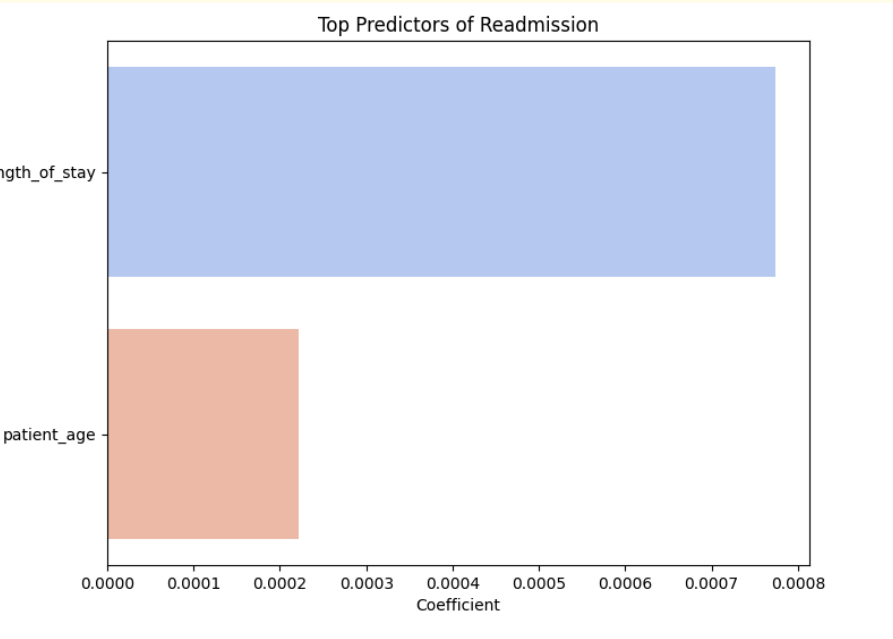
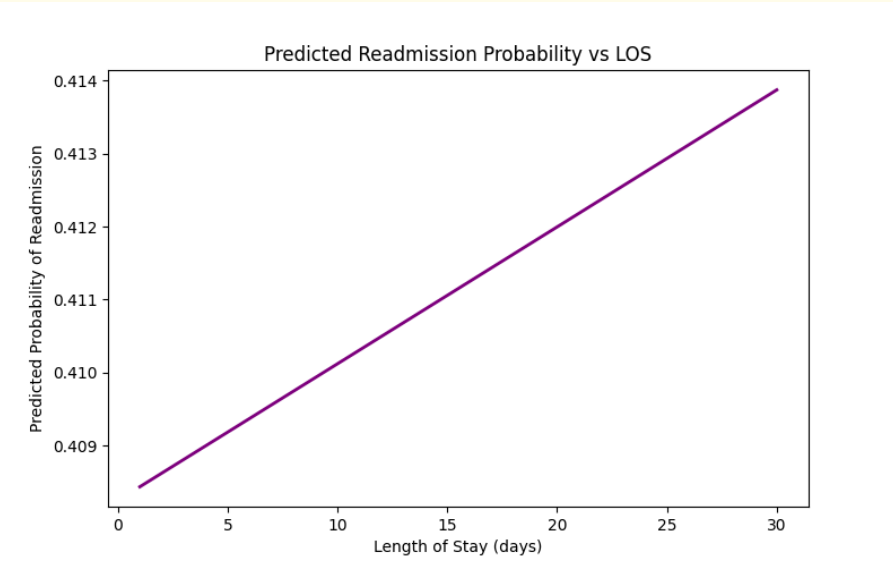
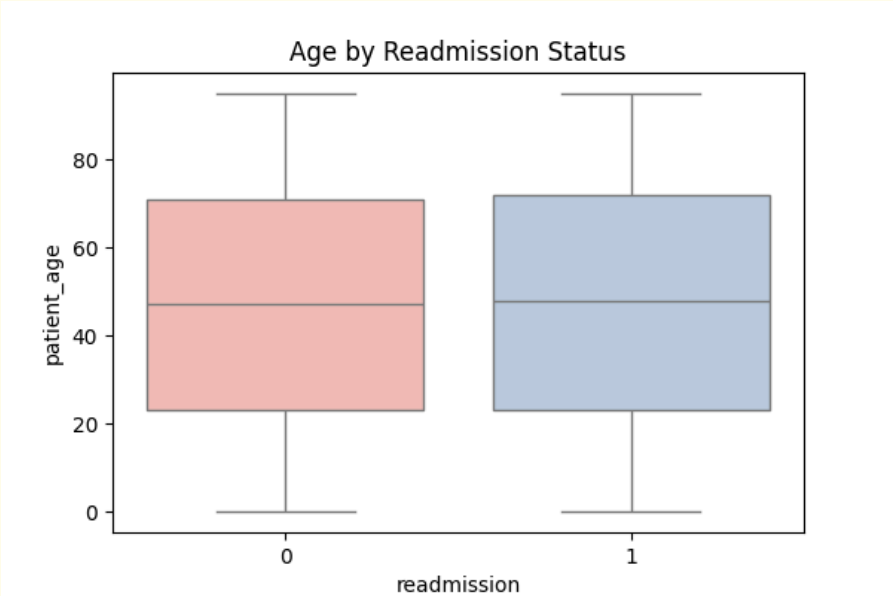
Finding 1: Length of stay does not significantly predict readmission.

- T-tests showed no meaningful difference between readmitted and non-readmitted patients.



Finding 2: Age and chronic respiratory diseases are strong predictors.

- Logistic regression coefficients indicate increased readmission likelihood for older adults and those with asthma or influenza.



Finding 3: Patient-centered discharge planning may mitigate risk.

- High-risk patients could benefit from tailored follow-up programs.

Conclusion and recommendations

This analysis highlights that patient-specific characteristics—especially age and disease type—are stronger predictors of hospital readmission than metrics like length of stay. Incorporating predictive analytics into discharge planning can help hospitals identify high-risk patients and allocate resources efficiently.